

Supplementary materials

Supplementary Table 1. Examination accuracy of (A) Examination 1 and (B) Examination 2 in Basic scenario

(A) Examination 1

		Observed PD State	
		SideEffect	$\neg(\text{SideEffect})$
True PD State	SideEffect	95%	5%
	$\neg(\text{SideEffect})$	25%	75%

(B) Examination 2

		Observed PD State	
		Fail	$\neg(\text{Fail})$
True PD State	Fail	95%	5%
	$\neg(\text{Fail})$	25%	75%

Supplementary Table 2. PK Assessment accuracy in Basic scenario

		Estimated PK State		
		Over	Optimal	Under
True PK State	Over	90%	10%	0%
	Optimal	5%	90%	5%
	Under	0%	10%	90%

Supplementary Table 3. PK Transition Probability when Protocol State is (A) Before Administration, (B) Regimen 0, (C) Regimen 1, or (D) Regimen 2.

(A) Before Administration

		post PK State		
		Optimal	Over	Under
present PK State	Baseline	93%	2%	5%

(B) Regimen 0

		post PK State		
		Optimal	Over	Under
present PK State	Optimal	99.7%	0.15%	0.15%
	Over	0.3%	99.7%	0%
	Under	0.3%	0%	99.7%

(C) Regimen 1

		post PK State		
		Optimal	Over	Under
present PK State	Optimal	99.7%	0.15%	0.15%
	Over	0.3%	99.7%	0%
	Under	0.3%	0%	99.7%

(D) Regimen 2

		post PK State		
		Optimal	Over	Under
present PK State	Optimal	99.7%	0.15%	0.15%
	Over	0.3%	99.7%	0%
	Under	0.3%	0%	99.7%

Supplementary Table 4. PK Transition Probability when the dosage regimen change (A) from Regimen 0 to Regimen 1, (B) from Regimen 1 to Regimen 2, (C) from Regimen 1 to Regimen 0, or (D) from Regimen 2 to Regimen 1.

(A) from Regimen 0 to Regimen 1

		post PK State		
		Optimal	Over	Under
present PK State	Optimal	10%	90%	0%
	Over	0%	100%	0%
	Under	90%	0%	10%

(B) from Regimen 1 to Regimen 2

		post PK State		
		Optimal	Over	Under
present PK State	Optimal	10%	90%	0%
	Over	0%	100%	0%
	Under	90%	0%	10%

(C) from Regimen 1 to Regimen 0

		post PK State		
		Optimal	Over	Under
present PK State	Optimal	10%	0%	90%
	Over	90%	10%	0%
	Under	0%	0%	100%

(D) from Regimen 2 to Regimen 1

		post PK State		
		Optimal	Over	Under
present PK State	Optimal	10%	0%	90%
	Over	90%	10%	0%
	Under	0%	0%	100%

Supplementary Table 5. PD Transition. Probability when present PK State is (A) Optimal, (B) Under, or (C) Over.

(A) Optimal

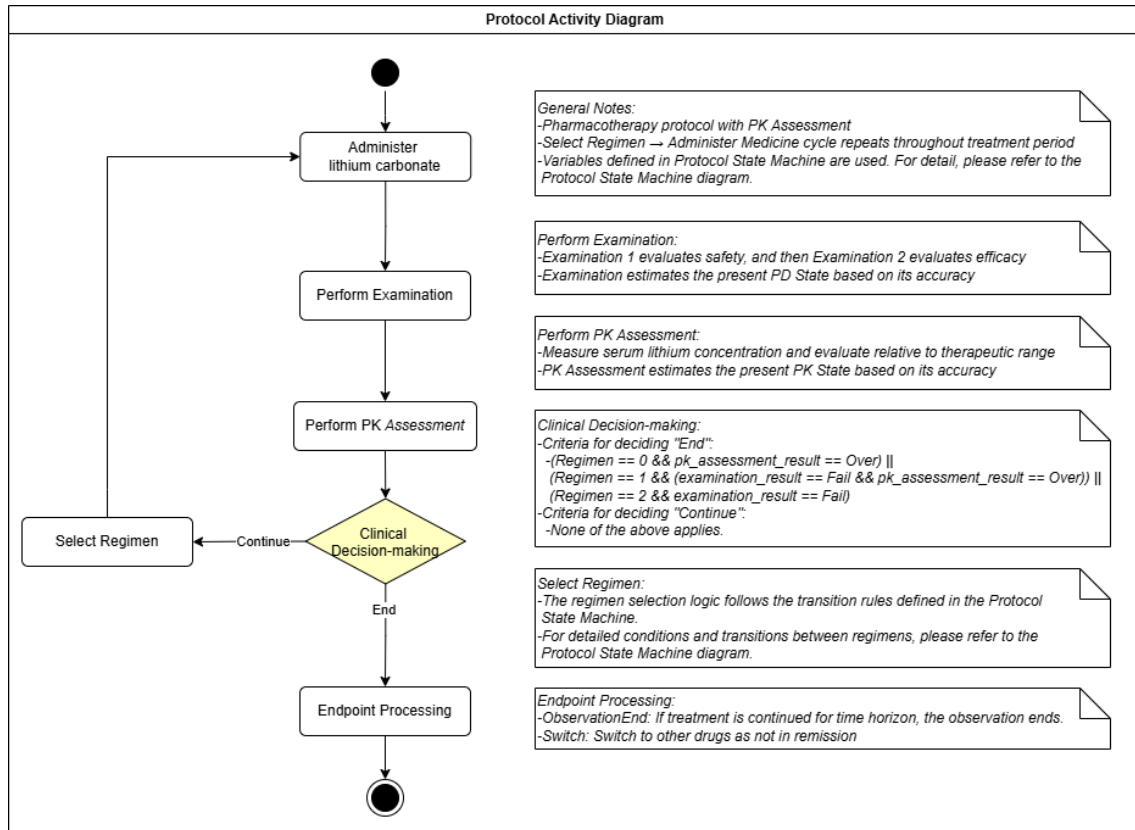
		post PD State		
		Improve	Fail	SE
present PD State	Baseline	98%	1%	1%
	Improve	99.8%	0.1%	0.1%
	Fail	99%	1%	0%
	SE	99%	0%	1%

(B) Under

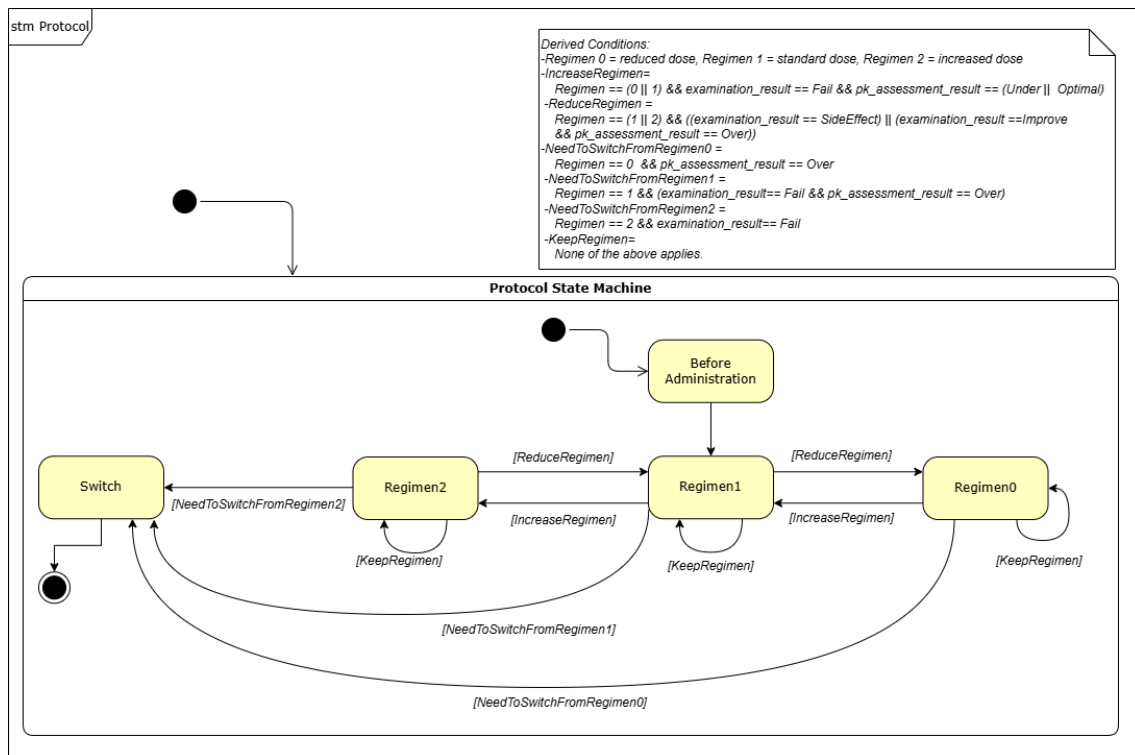
		post PD State		
		Improve	Fail	SE
present PD State	Baseline	10%	90%	0%
	Improve	10%	90%	0%
	Fail	0.2%	99.8%	0%
	SE	99%	0%	1%

(C) Over

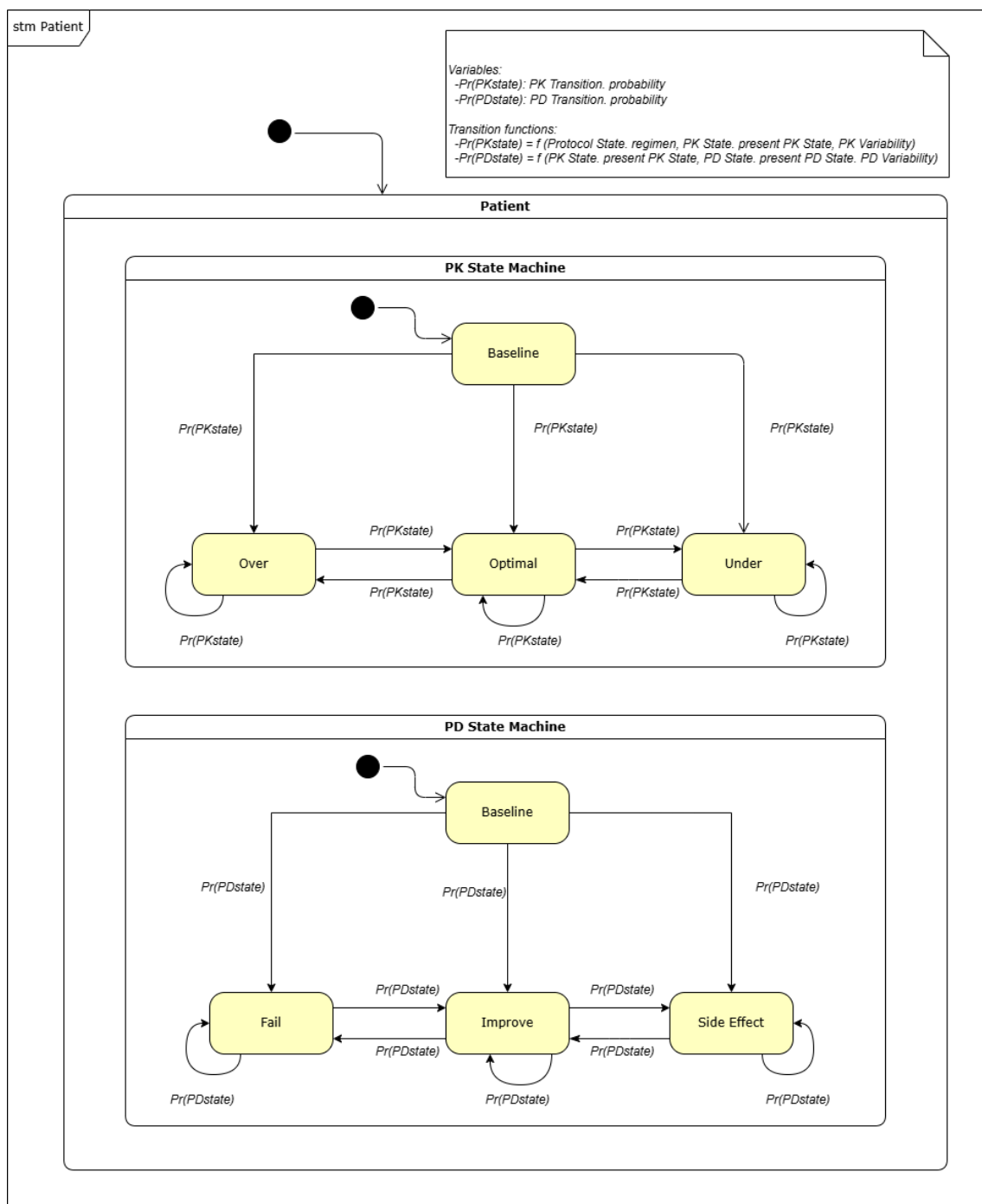
		post PD State		
		Improve	Fail	SE
present PD State	Baseline	10%	0%	90%
	Improve	10%	0%	90%
	Fail	99%	1%	0%
	SE	0.2%	0%	99.8%



Supplementary Fig. 1. Protocol activity diagram



Supplementary Fig. 2. Protocol state machine diagram



Supplementary Fig. 3. Patient state machine diagram